SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV
COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) Clearly interpret the scenario and explain the cause of low contrast.

(b) Identify all key issues affecting image visibility.

(c) Apply an appropriate enhancement technique to address the problem.

(d) Justify your choice with logical reasoning.

(e) Present your explanation in a clear and structured manner.

(a) Interpretation

A grayscale image appears dull because the difference between the object and background intensities is very small. Most pixel values are concentrated within a narrow intensity range, making it difficult to distinguish objects.

(b) Key Issues

- Poor visibility of objects.
- Loss of fine details.
- Low dynamic range of gray levels.
- Difficulty in segmentation and feature extraction.

(c) Enhancement Technique

Histogram Equalization

Histogram Equalization redistributes pixel intensities over the full gray-level range (0–255). It increases contrast by spreading frequently occurring intensity values.

(d) Justification

- Improves overall image contrast.
- Makes hidden details more visible.
- Suitable when the entire image has poor contrast.
- Easy to implement and computationally efficient.

(e) Conclusion

Histogram Equalization is the best choice because it enhances contrast, improves object visibility, and provides better image quality for further processing.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpret the scenario and explain the impact of uneven illumination.

(b) Identify key factors causing intensity variation.

(c) Apply a suitable enhancement method to normalize illumination.

(d) Justify your selected approach with reasoning.

(e) Provide a clear and well-structured explanation.

(a) Interpretation

The image contains bright and dark regions due to non-uniform lighting during image acquisition. This causes inconsistent brightness across the image.

(b) Key Issues

- Uneven intensity distribution.
- Some regions lose details due to overexposure.
- Dark regions hide important information.
- Reduced segmentation accuracy.

(c) Enhancement Technique

Adaptive Histogram Equalization (CLAHE)

CLAHE divides the image into small regions and enhances contrast locally instead of globally.

(d) Justification

- Corrects local brightness variations.
- Prevents over-enhancement of noise.
- Preserves image details in both bright and dark regions.
- Produces natural-looking images.

(e) Conclusion

CLAHE effectively normalizes illumination and improves visibility across the entire image.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise.

(b) Identify key issues impacting image quality.

(c) Apply an appropriate filtering technique.

(d) Justify why this method is more suitable than alternatives.

(e) Present your response clearly and logically.

(a) Interpretation

Salt-and-pepper noise appears as randomly distributed white (salt) and black (pepper) pixels caused by transmission errors or faulty sensors.

(b) Key Issues

- Reduced readability.
- Loss of image quality.
- False pixels interfere with processing.
- Difficult to identify actual image features.

(c) Filtering Technique

Median Filter

Each pixel is replaced by the median value of neighboring pixels.

(d) Justification

- Removes impulse noise effectively.
- Preserves image edges.
- Better than Mean Filter because it does not blur edges.
- Widely used for document restoration.

(e) Conclusion

Median Filtering is the most suitable method for removing salt-and-pepper noise while preserving important image information.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpret the scenario and explain why edges are blurred.

(b) Identify key issues related to smoothing and edge loss.

(c) Apply an appropriate technique to restore edges.

(d) Justify your decision with clear reasoning.

(e) Structure your explanation clearly.

(a) Interpretation

After smoothing, high-frequency components representing edges are reduced, causing object boundaries to become blurred.

(b) Key Issues

- Loss of edge sharpness.
- Reduced feature visibility.
- Poor object recognition.
- Lower segmentation accuracy.

(c) Enhancement Technique

Laplacian Sharpening

The Laplacian operator enhances rapid intensity changes corresponding to edges.

(d) Justification

- Restores edge sharpness.
- Highlights fine image details.
- Improves boundary visibility.
- Suitable after smoothing operations.

(e) Conclusion

Laplacian sharpening effectively restores lost details and improves image clarity.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpret the scenario and describe high-frequency noise.

(b) Identify the key issues affecting image clarity.

(c) Apply an appropriate frequency domain filtering method.

(d) Justify the choice of filter with reasoning.

(e) Present your answer in a clear and organized manner.

(a) Interpretation

The satellite image contains rapid random intensity variations that appear as fine-grained noise.

(b) Key Issues

- Reduced image clarity.
- Hidden features.
- Difficulty detecting objects.
- Poor image interpretation.

(c) Frequency Domain Technique

Gaussian Low-Pass Filter

The filter suppresses high-frequency components while preserving low-frequency information.

(d) Justification

- Smooths noise gradually.
- Produces fewer artifacts than Ideal Low-Pass Filter.
- Preserves overall image structure.
- Commonly used in satellite image processing.

(e) Conclusion

Gaussian Low-Pass Filtering effectively removes high-frequency noise while maintaining image quality.

6. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpret the scenario and explain the importance of boundary detection.

(b) Identify key challenges in detecting boundaries.

(c) Apply a suitable edge detection method.

(d) Justify your selection with logical reasoning.

(e) Present your explanation clearly and systematically.

(a) Interpretation

Medical images require accurate identification of tissue boundaries for diagnosis and treatment planning.

(b) Key Issues

- Weak edges.
- Presence of noise.
- Similar tissue intensities.
- Incomplete boundaries.

(c) Edge Detection Technique

Canny Edge Detection

It performs smoothing, gradient calculation, non-maximum suppression, and hysteresis thresholding.

(d) Justification

- High edge detection accuracy.
- Good noise suppression.
- Produces thin, continuous edges.
- Widely used in medical imaging.

(e) Conclusion

Canny Edge Detection provides reliable and accurate tissue boundary detection.

7. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

(a) Interpret the scenario and explain segmentation requirements.

(b) Identify key features enabling separation.

(c) Apply a suitable segmentation technique.

(d) Justify your decision logically.

(e) Present your response clearly.

(a) Interpretation

The object intensity is clearly different from the background, making separation easy.

(b) Key Features

- Large intensity difference.
- Uniform background.
- High object visibility.

(c) Segmentation Technique

Global Thresholding

A single threshold separates object pixels from background pixels.

(d) Justification

- Simple implementation.
- Fast execution.
- Accurate when intensity difference is large.
- Low computational complexity.

(e) Conclusion

Global Thresholding is the most suitable segmentation method for this situation.

8. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

(a) Interpret the scenario and explain segmentation challenges.

(b) Identify key issues such as intensity overlap.

(c) Apply an appropriate advanced segmentation method.

(d) Justify how the method handles complexity.

(e) Provide a structured and clear explanation.

(a) Interpretation

Different regions have overlapping intensity values, making intensity-based segmentation difficult.

(b) Key Issues

- Intensity overlap.
- Complex textures.
- Multiple objects.

- Noise interference.

(c) Segmentation Technique

Region Growing

Pixels are grouped based on intensity similarity and spatial connectivity.

(d) Justification

- Uses neighborhood information.
- Produces connected regions.
- Handles overlapping intensities better than thresholding.
- Gives more accurate segmentation.

(e) Conclusion

Region Growing effectively segments complex images with overlapping intensity values.

9. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

(a) Interpret the scenario and explain why edges are broken.

(b) Identify key issues affecting boundary continuity.

(c) Apply a suitable method to improve edge connectivity.

(d) Justify your approach logically.

(e) Present your explanation clearly.

(a) Interpretation

Detected object boundaries contain gaps because of noise or weak gradients.

(b) Key Issues

- Discontinuous boundaries.
- Incomplete objects.
- Poor shape representation.
- Difficult object recognition.

(c) Improvement Technique

Morphological Closing

Closing consists of Dilation followed by Erosion.

(d) Justification

- Connects broken edge segments.
- Fills small gaps.
- Produces continuous boundaries.
- Improves segmentation results.

(e) Conclusion

Morphological Closing improves edge continuity and enhances boundary detection.

10. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpret the scenario and define the combined requirement.

(b) Identify key issues involving noise and edge preservation.

(c) Apply an appropriate sequence of techniques.

(d) Justify the order and selection with strong reasoning.

(e) Present your response in a clear and structured format.

(a) Interpretation

The inspection system must remove noise while preserving important edges to accurately detect defects.

(b) Key Issues

- Noise hides defects.
- Excessive smoothing removes edges.
- Sharp boundaries are essential.
- High detection accuracy is required.

(c) Sequence of Techniques

Step 1: Apply **Median Filter** to remove noise.

Step 2: Apply **Canny Edge Detection** (or Laplacian Sharpening) to detect and enhance edges.

(d) Justification

- Median Filter removes impulse noise without blurring edges.
- Canny accurately detects defect boundaries after noise removal.
- The sequence ensures clean images with preserved edge information.

- Improves inspection accuracy and reliability.

(e) Conclusion

Using a **Median Filter followed by Canny Edge Detection** provides effective noise removal while preserving important edges, making it ideal for industrial defect detection.

Code of Conduct:

*I **KANMANI K B (192524306)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand